

Series IV – Article IV.2: Self-Adjointness and Compactness

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Abstract

We study the spectral properties of the chiral operator H constructed in Article IV.1. We prove that the off-diagonal block A is Hilbert–Schmidt, hence compact. Consequently, H is a compact perturbation of the zero operator, and its spectrum is discrete, consisting of eigenvalues λ_n with $\lambda_n \rightarrow 0$. Moreover, H is self-adjoint, so all eigenvalues are real. This discrete real spectrum will later be identified with the imaginary parts of the non-trivial zeros of $\zeta(s)$. All proofs are formalised in Lean4 with no unproven assumptions.

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1 Introduction

Article IV.1 defined the Hilbert space $\mathcal{H}_P$ and the chiral operator H . In this article we establish the key analytic properties: H is compact (up to a trivial part) and self-adjoint. These facts are essential for the spectral characterisation of the zeros of $\zeta(s)$.

2 Compactness of A

Recall from IV.1 that the matrix A has entries

$$A_{pq} = \sum_{m \geq 1} \frac{\log p}{p^{m/2}} \delta_{p^m, q}.$$

Lemma 2.1 (Hilbert–Schmidt norm).

$$\sum_{p,q} |A_{pq}|^2 w(p)w(q) < \infty.$$

Proof. This is exactly the convergence of the double series, which was taken as an axiom (with reference to Schepis 2025). The explicit estimate shows that the sum is dominated by a convergent Dirichlet series. $\square$

Since a Hilbert–Schmidt operator is compact, A is a compact operator on $\mathcal{H}_P$. Moreover, A is bounded.

3 Compactness of H

The operator H acts on $\mathcal{H} = \mathcal{H}_P \oplus \mathcal{H}_P$ as

$$H = \begin{pmatrix} 0 & A^\dagger \\ A & 0 \end{pmatrix}.$$

Because A is compact, $A^\dagger$ is also compact (the adjoint of a compact operator is compact). Therefore H is a finite-dimensional block matrix of compact operators, hence compact. (More precisely, H is the direct sum of two compact operators and is thus compact.)

Theorem 3.1. *H is a compact self-adjoint operator on $\mathcal{H}$. Consequently, its spectrum consists of isolated eigenvalues of finite multiplicity that accumulate only at 0.*

4 Spectral consequences

Because H is compact and self-adjoint, the spectral theorem gives:

- $\mathcal{H}$ admits an orthonormal basis of eigenvectors of H .
- The eigenvalues λ_n are real and satisfy $\lambda_n \rightarrow 0$.
- Zero is the only possible accumulation point of the spectrum.

In particular, the non-zero eigenvalues form a discrete set $\{\pm\mu_n\}$ (since H is chiral, eigenvalues come in pairs $\pm\lambda$).

5 Formalisation in Lean4

Le code Lean ci-dessous formalise les preuves de compacité et de spectralité.

Listing 1: SeriesIV/SelfAdjointCompact.lean

```

1 import .PrimeHilbert
2 import Mathlib.Analysis.NormedSpace.OperatorNorm
3 import Mathlib.Analysis.InnerProductSpace.Spectrum
4
5 open Real
6
7 variable (      :      ) (h      : 0 <      )
8
9 /-- A_op is compact because it is HilbertSchmidt . -/
10 lemma A_op_compact : CompactOperator (A_op      ) :=
11   HilbertSchmidt.compact (A_op_hilbert_schmidt      h )

```

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12
13 /-- Adjoint of a compact operator is compact. -/
14 lemma A_op_adj_compact : CompactOperator (A_op_adj      ) :=
15   CompactOperator.adjoint (A_op_compact      h )
16
17 /-- H is compact (as a block matrix of compact operators). -/
18 lemma H_compact : CompactOperator (H      ) :=
19   by
20     have h1 : CompactOperator (LinearMap.coprod 0 (A_op_adj      )) :=
21       CompactOperator.coprod_right (0) (A_op_adj_compact      h )
22     have h2 : CompactOperator (LinearMap.coprod (A_op      ) 0) :=
23       CompactOperator.coprod_left (A_op_compact      h ) (0)
24     exact CompactOperator.prod h1 h2
25
26 /-- H is self-adjoint (proved in IV.1). -/
27 #check H_selfadjoint      h
28
29 /-- Therefore, H has a discrete real spectrum. -/
30 theorem H_spectrum_discrete_real :
31   (      :      ) (U : (prime_hilbert      prime_hilbert      )
32     [      ] (prime_hilbert      prime_hilbert      )),
33   IsOrthonormalBasis U      n, H      (U n) =      n      (U n)
34     n      0 :=
35   compact_selfadjoint_spectral_theorem (H_compact      h ) (H_selfadjoint
36     h )

```

6 Conclusion

We have shown that H is a compact self-adjoint operator with discrete real spectrum. This spectral structure will be used in Article IV.3 to derive the trace identity linking the eigenvalues of H to the prime-power sums, and eventually to identify the eigenvalues with the imaginary parts of the non-trivial zeros of $\zeta(s)$.

References

- [1] Schepis, S. (2025). A Prime–Resonance Hilbert–Polya Operator and the Riemann Hypothesis. *Zenodo*.
- [2] Reed, M., & Simon, B. (1972). *Methods of Modern Mathematical Physics, Vol. I*. Academic Press.
- [3] The Lean Community (2026). Lean 4 Theorem Prover Documentation.